RACE
Regular Analysis Through Continuous Exercise

Surface Chemistry : RACE-01

Adsorption

1. Physical adsorption is appreciable at

(A) higher temperature (B) lower temperature

(C) at room temperature (D) 100°C

CSF130

2. Which of the following is not a characteristic of chemisorption?

(A) Adsorption is irreversible.

(B) Adsorption decreases with increase in temperature.

(C) Adsorption is specific.

(D) Adsorption increases with increase in surface area.

CSF131

3. Which gas will be adsorbed on a solid to greater extent?

(A) A gas having non-polar molecules. (B) A gas having higher critical temperature.

(C) A gas having lowest critical temperature. (D) A gas having higher critical pressure.

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4. The volumes of gases H_2, CH_4, CO_2 and NH_3 adsorbed by 1 g of charcoal at 288K are in the order

(A) $H_2 > CH_4 > CO_2 > NH_3$ (B) $CH_4 > CO_2 > NH_3 > H_2$

(C) $CO_2 > NH_3 > H_2 > CH_4$ (D) $NH_3 > CO_2 > CH_4 > H_2$

CSF135

5. The enthalpy of adsorption in physisorption lies in the range of

(A) (40 – 400) kJ/mol (B) (80 – 240) kJ/mol

(C) (20 – 40) kJ/mol (D) (1 – 10) kJ/mol

CSF136

6. Graph between $\log\left(\frac{x}{m}\right)$ and log P is a straight line at an angle 45° with intercept on y-axis, 0.3010.

The amount (in g) of the gas absorbed per g of the adsorbent when pressure is 0.2 atm is (assume that the adsorption obey Freundlich isotherm)

(A) 0.4 (B) 0.6 (C) 0.8 (D) 0.2

CSF161

7. Desorption of a gas from metal surface follows first-order kinetics. The rate constant of desorption can be given by Arrhenius equation. If the desorption of hydrogen on manganese is found to increases 10 times on increasing the temperature from 600 to1000 K, the activation energy of desorption is (ln 10 = 2.3)

(A) 6.0 kcal/mol (B) 6.9 kcal/mol

(C) 3.0 kcal/mol (D) 57.4 kcal/mol

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8. According to Freundlich adsorption isotherm, which of the following is correct?

(A) $\frac{x}{m} \propto P^{1/n}$

(B) $\frac{x}{m} \propto P^{0}$

(C) $\frac{x}{m} \propto P^{1}$

(D) All the above are correct for different ranges of pressure.

CSF163

9. At 70 K, the adsorption of $N_2(g)$ at iron surface obeys Freundlich adsorption isotherm. The following data is collected experimentally.

P (/bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

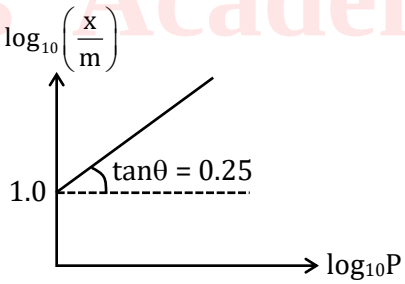
Here, $\frac{x}{m}$ is the mass (in g) of $N_2(g)$ adsorbed per g of iron at P bar pressure. The moles of $N_2(g)$ adsorbed per g iron at 36 bar and 70 K is

(A) $\frac{6}{10}$ (B) $\frac{3}{140}$ (C) $\frac{3}{70}$ (D) $\frac{3}{280}$

CSF164

10. Adsorption of hydrogen gas over palladium surface obey Freundlich's adsorption isotherm as shown in the graph. The mass (in g) of hydrogen gas adsorbed on the surface of 1.0 g palladium at a pressure of 8.1×10^{-3} atm is

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11. Nitrogen gas adsorbed on charcoal to the extent of 0.387 cm^3/g at a pressure of 1.6 atm and at temperature of 200 K, but at 250 K the same amount of adsorption was achieved only when the pressure was increased to 32 atm. The magnitude of molar enthalpy of adsorption (in kcal/mol) of nitrogen on charcoal is (ln 20 = 3.0)

CSF158

12. Twenty per cent of surface sites are occupied by N_2 molecules. The density of surface site is $6.023 \times 1014\ cm^{-2}$ and the total surface area is 1000 cm^2. The catalyst is heated to 300 K while N_2 is completely desorbed into a pressure of 0.001 atm and volume of 2.46 cm^3. The number of active sites occupied by each N_2 molecule is

CSF190

Catalysis

13. Efficiency of the catalyst depends on its

(A) molecular mass (B) number of free valencies

(C) physical state (D) amount used

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14. The following information is available for the reaction R P.

$R \longrightarrow P$: rate = r_1

$R \xrightarrow{A} P$: rate = r_2

$R \xrightarrow{B} P$: rate = r_3

$R \xrightarrow{C} P$: rate = r_4

$R \xrightarrow{D} P$: rate = r_5

$R \xrightarrow{A+C} P$: rate = r_6

$R \xrightarrow{A+D} P$: rate = r_7

If $r_3 < r_1 = r_4 = r_5 < r_7 < r_2 < r_6$, then the only incorrect statement is

(A) B is an inhibitor. (B) C is a catalytic promoter of catalyst A.

(C) D is a catalytic poison of catalyst A. (D) A is a catalytic promoter of catalyst C.

CSF165

15. Which of the following statements is false for catalyst?

(A) A catalyst initiates the reaction.

(B) It does not alter the position of equilibrium in a reversible reaction.

(C) A catalyst remains unchanged in quantity and composition at the end of reaction.

(D) Catalysts are sometimes very specific with respect to a reaction.

CSF168

16. Regarding criteria of catalyst, which one of the following statements is not true?

(A) The catalyst is unchanged chemically after the reaction.

(B) A small quantity of catalyst is often sufficient to bring about a considerable amount of the reaction.

(C) In reversible reaction, the catalyst alters the equilibrium position.

(D) The catalyst alters the rate of reaction.

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	B	B	D	C	A	B	A	B	3
Question	11	12	13	14	15	16				
Answer	6	2	B	D	A	C				

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Regular Analysis Through Continuous Exercise

Surface Chemistry : RACE-02

Colloids and Classification of Colloids

1. On adding $AgNO_3$ solution into KI solution, a negatively charged colloidal sol is obtained when they are in
(A) 100 ml of 0.1 M $AgNO_3$ + 100 ml of 0.1 M KI
(B) 100 ml of 0.1 M $AgNO_3$ + 50 ml of 0.2 M KI
(C) 100 ml of 0.1 M $AgNO_3$ + 50 ml of 0.1 M KI
(D) 100 ml of 0.1 M $AgNO_3$ + 100 ml of 0.15 M KI

CSF150

2. Among the following colloids, the number of colloids having negatively charged colloidal particles is: Soap solution of sodium palmitate, sol of AgCl obtained on dissolving excess $AgNO_3$ into KCl solution, gold sol, $Fe(OH)_3$ sol, silicic acid sol, basic dye, acidic dye, metal sulphide sol, sol of AgCl obtained by dissolving excess KCl in $AgNO_3$ solution.

CSF157

3. The minimum number of phases in a colloidal system is

CSF192

Properties of Colloids

4. Bleeding is stopped by the application of ferric chloride. This is because
(A) the blood starts flowing in opposite direction.
(B) the ferric chloride seals the blood vessel.
(C) the blood reacts and forms a solid which seals the blood vessel.
(D) the blood is coagulated and thus, the blood vessel is sealed.

CSF145

5. Gold number is a measure of
(A) the amount of gold present in the colloidal solution.
(B) the amount of gold required to break the colloid.
(C) the amount of gold required to protect the colloid.
(D) the protective power of lyophilic colloid.

CSF146

6. The gold numbers of A, B, C and D are 0.04, 0.002, 10 and 25, respectively. The protective powers of A, B, C and D, respectively, are in the order
(A) A > B > C > D (B) B > A > C > D (C) D > C > B > A (D) C > A > B > D

CSF147

7. Which of the following substances will have least CMC value?
(A) $C_6H_{13}COOK$ (B) $C_9H_{19}SO_3Na$ (C) $C_{11}H_{23}NH_4Cl$ (D) $C_8H_{17}COONa$

CSF152

8. Which of the following is not a property of hydrophilic sol?
(A) High concentration of dispersed phase can be easily attained.
(B) Coagulation is reversible.
(C) Viscosity and surface tension are nearly as that of water.
(D) The charge on the particles may depend on the pH value and it may be positive, negative or even zero.

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9. A quantity of 1.9×10^{-4} g of the metal having density 19 g/ml is dispersed in 1 L of water to give a sol having spherical metal particles of radius 10 nm. The approximate number of metal sol particles per cm^3 of the sol is

(A) 2.39×10^9 (B) 4×10^{10} (C) 1.9×10^9 (D) 2.8×10^{-12}

CSF184

10. Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient conditions is

(A) $CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$ (B) $CH_3(CH_2)_{11}OSO_3^-Na^+$

(C) $CH_3(CH_2)_6COO^-Na^+$ (D) $CH_3(CH_2)_{11}N^+(CH_3)_3Br^-$

CSF185

11. Among the electrolytes Na_2SO_4, $CaCl_2$, $Al_2(SO_4)_3$ and NH_4Cl, the most effective coagulating agent for Sb_2S_3 is

(A) Na_2SO_4 (B) $CaCl_2$ (C) $Al_2(SO_4)_3$ (D) NH_4Cl

CSF186

12. Match the columns.

Column I	**Column II**
(1) Coagulation	(P) Scattering of light
(2) Peptization	(Q) Purification of colloidal solution
(3) Tyndall effect	(R) Addition of an electrolyte
(4) Dialysis	(S) Precipitation of colloidal solution

(A) 1-P; 2-Q; 3-R; 4-S (B) 1-S; 2-R; 3-P; 4-Q (C) 1-R; 2-S; 3-Q; 4-P (D) 1-Q; 2-R; 3-P; 4-S

CSF188

13. Molecular formula of starch can be represented as $(C_6H_{10}O_5)_n$. If the gold number of one such starch is 6.48 and 0.01 millimoles of this starch is required to be added in 10 ml of red gold sol to just prevent coagulation by adding 1 ml of 10% sodium chloride solution, then the value of 'n' is

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Preparation of Colloids

14. Which metal sol (dispersion medium being water) cannot be prepared by Bredig's arc method?

(A) K (B) Cu (C) Au (D) Pt

CSF176

15. The formed colloid in following reaction is SnO_2 + HCl (Excess)

(A) $SnCl_4/Cl^-$ (B) $SnCl_4/O^{2-}$ (C) $SnCl_4/H^+$ (D) $SnCl_4/Sn^{4+}$

CSF177

16. Which of the following reaction is not used in the preparation of colloidal solution?

(A) $2H_2S + SO_2 \rightarrow 3S + 2H_2O$ (B) $2Mg + CO_2 \rightarrow 2MgO + C$

(C) $FeCl_3 + 3H_2O \rightarrow Fe(OH)_3 + 3HCl$ (D) $2AuCl_3 + 3SnCl_2 \rightarrow 2Au + 3SnCl_4$

CSF183

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	6	2	D	D	B	C	C	A	A
Question	11	12	13	14	15	16				
Answer	C	B	4	A	A	B				

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Important Notes

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